

Weizheng Bai

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EDUCATION

M.Sc.of Civil and Hydraulic Engineering

Sep.2022-June.2025

University of Chinese Academy of Sciences, Beijing, China

GPA: 3.69/4.0

Supervisor: Prof. Xilin Shi

Non-Degree Coursework., Civil and Hydraulic Engineering

Sep.2022-June.2023

Wuhan University, Wuhan, China

B.Sc.of Civil Engineering

Sep.2017-June.2021

Shandong University of Science and Technology, Qingdao, China

GPA: 3.01/4.0

RESEARCH INTERESTS

- Underground hydrogen storage
- Hydrogen seepage
- The application of artificial intelligence in underground energy storage

SKILLS

- **Software**
COMSOL Multiphysics, FLAC 3D, Origin, Microsoft office
- **Programming Skills**
Matlab
- **Language**
Chinese(native), English(IELTS:6.5)
- **Interests**
Fitness, Soccer, Climbing, Singing

PUBLICATIONS

- **Bai W**, Shi X*, Yang C, et al. Assessment of the potential of salt mines for renewable energy peaking in China[J]. *Energy*, 2024, 300: 131577. (JCR Q1, IF=9.4)
 - **Bai W**, Lu J, Wang J, et al. Numerical Simulation Study of Salt Cavern CO2 Storage in Power-to-Gas System[J]. *Energies*, 2024, 17(22): 5786. (JCR Q3, IF=3.2)
 - **Bai W**, Shi X*, Zhu S, et al. Opportunities and challenges of large-scale salt cavern hydrogen storage in China coupled with renewable energy sources[J]. *Clean Energy*, 2025, 9(1): 179-196. (JCR Q3, IF=3.7)
 - **Bai W**, Shi X*, Zhu S, Wei X, Huang Y. Leakage risk assessment system for salt cavern hydrogen storage[J]. *Deep Undergr SciEng*. 2025;1-14.(JCR Q1, IF=5)
 - Wei X, Shi X*, Ma H, **Bai W**, et al. Oil storage and debrining process in insoluble sediment voids for underground salt cavern energy storage: An experimental study[J]. *Geoenergy Science and Engineering*, 2024, 243: 213359.
 - Wang J, Li P, **Bai W**, et al. Mechanical Behavior of Sediment-Type High-Impurity Salt Cavern Gas Storage during Long-Term Operation. *Energies*, 2024, 17(16): 3983.
 - Zhu S, Shi X, Yang C, **Bai W**, et al. Site selection evaluation for salt cavern hydrogen storage in China[J]. *Renewable Energy*, 2024, 224: 120143.
 - Liu X, Huang Y, Shi X, **Bai W**, et al. Offshore Wind Power—Seawater Electrolysis—Salt Cavern Hydrogen Storage Coupling System: Potential and Challenges[J]. *Energies*, 2025, 18(1): 169.
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RESEARCH EXPERIENCES

- **Research and development of key technologies for underground hydrogen storage** 01/2023-12/2024
 - a) To carry out research on the transport patterns of hydrogen in the surrounding rock of underground hydrogen storage.
 - b) Establishment of a system for evaluating the suitability of geological conditions for underground hydrogen storage.
 - c) Investigated the distribution and geological data of hydrogen storage geological bodies in China, and screen feasible sites for building underground hydrogen storage.
- **National Key R&D Program of China: Applicability and key safety technologies of geological hydrogen storage facilities** 12/2024-11/2027

To establish an evaluation index system for the sealing and stability of geological bodies. Based on data such as geological structure, surrounding rock properties, chamber morphology, stress field distribution, formation fluids and surrounding environment, propose methods for evaluating the applicability of hydrogen storage in geological bodies and selecting favorable blocks.

AWARDS

- China National Scholarship(TOP 1%) 2024
- Outstanding Graduate of University of Chinese Academy of Sciences 2025
- Outstanding Student of the University of Chinese Academy of Sciences 2024
- First-class Award Scholarship of the Graduate School, UCAS 2022-2024
- Third Prize in the National College Students' Mathematics Competition 2020
- First-class award in National English Competition for college Students 2019

Reviewer Activities

- Tunnelling and Underground Space Technology
- Energy
- Fuel
- Energy Strategy Reviews
- Journal of Rock Mechanics and Geotechnical Engineering
- Journal of Energy Storage
- Sustainable Energy Technologies and Assessments
- Case Studies in Thermal Engineering
- Unconventional Resources